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SHOCK MOUNTED LINEAR SUSPENSION HEIGHT SENSOR WITH ACCELEROMETER

Abstract

A sensor assembly for a suspension assembly of a vehicle may include a sensor for detecting displacement data of a target magnet and a sensor casing operably coupled to an external body of a shock absorber of the suspension assembly and the sensor. The sensor may include a linear flexible printed circuit that may be disposed along a longitudinal axis of the shock absorber to measure displacement data of the target magnet relative to the shock absorber to determine the suspension assembly position. The sensor assembly may further include a single axis accelerometer measuring rate of change of suspension velocity data in a direction along the longitudinal axis of the shock absorber, and the target magnet operably coupled to the suspension assembly via a suspension assembly component other than the external body of the shock absorber.

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Background/Summary

TECHNICAL FIELD

[0001] Example embodiments generally relate to vehicle suspension and, more particularly, relate to a suspension system with an integrated shock and suspension height sensor with an accelerometer.

BACKGROUND

[0002] Vehicles commonly employ an independent suspension that allows each wheel to move relative to the vehicle chassis independent of the other wheels. The components and geometries used for independent suspension designs can vary to some degree. However, a typical independent suspension system will employ dampers or shock absorbers (or simply “shocks”) that are designed to provide damping for pitch (i.e., oscillation about a lateral axis of the vehicle), heave and roll, as well as individual wheel contacts. The shocks generally resist compression and rebound with damping forces that are applied over a range of travel of a piston rod.

[0003] The shocks selected for a particular vehicle are generally chosen based on the expectation of normal vehicle dynamics scenarios that are encountered during routine driving conditions. Some typical suspension components that may be considered for selection include springs and dampers (e.g., gas shocks). Moreover, some shocks employ what is referred to as a coil-over design in which a coil spring is provided coaxial with the damper such that the damper is inside the coil spring. Many modern vehicles utilize semi-active or active suspensions to continuously modify damping levels of a vehicle's suspension system using shock absorbers. To accurately modify damping and identify shock absorber condition, wheel position and rate of change of suspension velocity are needed.

[0004] Thus, it may be desirable to be able to include an easily installed sensor assembly to accurately and quickly measure wheel position and rate of change of suspension velocity.

BRIEF SUMMARY OF SOME EXAMPLES

[0005] In accordance with an example embodiment, a sensor assembly for a suspension assembly of a vehicle may be provided. The sensor assembly may include a sensor for detecting displacement data of a target magnet and a sensor casing operably coupled to an external body of a shock absorber of the suspension assembly and the sensor. The sensor may include a linear flexible printed circuit that may be disposed along a longitudinal axis of the shock absorber to measure displacement data of the target magnet relative to the shock absorber to determine the suspension assembly position. The sensor assembly may further include a single axis accelerometer measuring rate of change of suspension velocity data in a direction along the longitudinal axis of the shock absorber, and the target magnet operably coupled to the suspension assembly via a suspension assembly component other than the external body of the shock absorber.

[0006] In yet another example embodiment, a suspension system for a vehicle may be provided. The suspension assembly may include a shock absorber operably coupled to a wheel assembly and a chassis of the vehicle, and a sensor assembly operably coupled to the shock absorber. The sensor assembly may include a sensor for detecting displacement data of a target magnet, a sensor casing operably coupled to an external body of the shock absorber of the suspension assembly and the sensor. The sensor may include a linear flexible printed circuit that may be disposed along a longitudinal axis of the shock absorber to measure displacement data of the target magnet relative to the shock absorber to determine the suspension assembly position. The sensor assembly further include a single axis accelerometer measuring rate of change of suspension velocity data in a direction along the longitudinal axis of the shock absorber, and the target magnet operably coupled to the suspension assembly via a suspension assembly component other than the external body of the shock absorber.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

[0007] Having thus described the invention in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0008] FIG. **1** illustrates a block diagram of a portion of a vehicle suspension system in accordance with an example embodiment;

[0009] FIG. **2** shows a block diagram of internal components in a portion of a vehicle suspension system in accordance with an example embodiment;

[0010] FIG. **3** illustrates a perspective view of a vehicle suspension system in accordance with an example embodiment;

[0011] FIG. **4** shows an isolated perspective view of a sensor assembly in accordance with an example embodiment;

[0012] FIG. **5** illustrates an isolated perspective view of a rebound stop for a piston in accordance with an example embodiment; and

[0013] FIG. **6** shows an isolated perspective view of a shock absorber outer body in accordance with an example embodiment.

DETAILED DESCRIPTION

[0014] Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed, the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.

[0015] Some example embodiments described herein may address the issues described above. In this regard, for example, some embodiments may provide a more efficient shock or suspension assembly design that employs position sensors that are configured to measure ride height into existing components. Some embodiments may provide for a target magnet that may be installed on a first suspension component (e.g., a piston assembly), and a position sensor (e.g., in the form of an elongated printed circuit board (PCB) with wiring assemblies thereon) that is provided on another suspension component that moves in a linear fashion (e.g., an external body of a shock absorber). As a result, movement of the first and second suspension components relative to each other (displacement) may be measured and converted into information on ride height (position) based on such movement to effectively instantiate an integrated ride height sensor that is relatively simple in construction, but also accurate in operation.

[0016] FIG. **1** illustrates a block diagram of a suspension assembly **100** employing a shock absorber **110**. FIG. **2** shows a block diagram of some internal components of the shock absorber **110** of a vehicle suspension system. The suspension assembly **100** may employ the shock absorber **110** to dampen jounce and rebound forces that may be initiated between a body or chassis **120** of the vehicle and a wheel **130** (or wheel assembly components). In some cases, the chassis may include or be defined by a frame, and the frame may additionally be formed of one or more casted subframes. In this regard, the shock absorber **110** may operably couple the chassis **120** to the wheel **130**. The shock absorber **110** may include a strut module or damper tube **140** with a rod **142** that extends into the damper tube **140**. The damper tube **140** may be operably coupled to the wheel **130** via a first mechanical interface **144**. The rod **142** may be operably coupled to the body chassis **120** via a second mechanical interface **146**. In some cases, the first mechanical interface **144** and the second mechanical interface **146** may be a ball and socket, a screw, a nut and bolt, or any other interface that securely operably couples the components. The rod **142** may also be operably

coupled to the damper tube **140** in such a way as to resist movement of the rod **142** via gas, fluid, springs, or other damping media that may be disposed within the damper tube **140**.

[0017] The rod **142** may be a component of a piston assembly. The piston assembly may include a piston head **180**. The piston head **180** may be operably coupled to the rod **142** and the damper tube **140**. The piston head **180** may have a variety of constructions or shapes depending on the shock absorber **110**. In some cases, the piston head **180** may have a plurality of apertures to allow damping media to travel through the piston head **180**. In an example embodiment, the piston head **180** may form a complete seal between an internal wall of the damper tube **140** and the external edge of the piston head **180** with the damper tube **140**.

[0018] As the suspension assembly **100** encounters compression and extension due to jounce and rebound associated with traversing uneven surfaces, the rod **142** may be plunged further into the damper tube **140** and alternately extend farther out of the damper tube **140**, respectively, relative to the position shown in FIG. **1**. In some cases, both the damper tube **140** and the rod **142** may move relative to one another in a movement direction represented by the double arrow **165**. By measuring the displacement along the movement direction, it may be possible to determine ride height or position for the corresponding wheel or corner of a vehicle employing the suspension assembly **100**.

[0019] As noted above, some complicated designs are already in place to try to measure this movement. However, such designs add many parts to those generally shown in FIG. **1**, and are therefore complicated and inefficient to employ. Moreover, many such designs do not provide linear outputs that are reliably accurate. Meanwhile, example embodiments may integrate measurement components into the components shown in FIG. **1** to provide quick and accurate ride height measurement for the suspension assembly **100**.

[0020] In this regard, as shown in FIG. **1**, a sensor assembly **200** may be installed in or attached to the shock absorber **110**. The sensor assembly **200** may include a sensor casing **210** and a linear sensor **220**. The sensor casing **210** may operably couple the linear sensor **220** to an external body **111** of the shock absorber **110**. In some cases, if the shock absorber **110** is a single tube shock absorber, the external body **111** may be specifically the external side of the damper tube **140**. In an example embodiment, a target magnet **155** may, for example, generate a magnetic field that is of a known magnitude, and at a known location (i.e., at the corresponding portion of the shock absorber **110**, and at a fixed position relative to a component of the suspension assembly **100**). In some cases, the target magnet **155** may be within the damper tube **140** or may be operably coupled to an external component of the suspension assembly **100**. The target magnet **155** may be any device or object that generates a known magnetic field. In an example embodiment, the target magnet **155** may be operably coupled to any component of the suspension assembly, as long as its position may be known. In some cases, a rebound stop **150** may be operably coupled to the piston head **180** to help assist in force control within the shock absorber **110**, and specifically the damper tube **140**. In an example embodiment, the rebound stop **150** may include the target magnet **155** for the linear sensor **220** of the sensor assembly **200**. The inclusion of the target magnet **155** within the rebound stop **150** may allow for assembly of the sensor assembly **200** onto the shock absorber **110** without altering piston dynamics. In this regards, the shock absorber **110** calibration may not be affected and may allow for calibration and tuning of the shock absorber **110** prior to sensor assembly **200** installation and shock absorber **110** installation.

[0021] Meanwhile, the linear sensor **220** may be disposed along a length of the external body **111**. In some cases, the linear sensor **220** may be disposed along a longitudinal axis **190** of the shock absorber **110**. In an example embodiment, the linear sensor **220** may be disposed along longitudinal axis **190** of the external body **111** of the shock absorber **110**. As the rod **142** (and the piston head **180**, the rebound stop **150**, and thus the target magnet **155**) may move up or down responsive to jounce and rebound, the location of the target magnet **155** correspondingly moves relative to the linear sensor **220**. In some cases, as the external body **111** (and the damper tube **140**) may move up

or down responsive to jounce and rebound, the location of the linear sensor **220** correspondingly moves relative to the target magnet **155**. The linear sensor **220** may be configured to track such movement and ride height information may correspondingly be determined based on such movement.

[0022] In an example embodiment, the linear sensor **220** may be embodied as and may include an elongated PCB **222** that extends along a surface of the external body **111**, as shown in FIG. 3. Moreover, in some cases, the PCB **222** may include, carry or be in communication with a wiring assembly or connector assembly **230**. The connector assembly **230** may be a traditional over-molded integrated circuit that may include various connectors, logic circuits, additional sensors, and transceivers. The connector assembly **230** may have a current run therethrough, and the current may generate a corresponding magnetic field. The magnetic field generated by the connector assembly **230** may be interacted with by the magnetic field of the target magnet **155**. For example, the target magnet **155** may distort the magnetic field generated by the connector assembly **230**. The distortion or modification of the magnetic field of the connector assembly **230** may be measured/monitored by the PCB **222** of the linear sensor **220**, and a corresponding ride height information determination may be made by the PCB **222** (or another processing component operably coupled thereto) based on the displacement or position (displacement data) of the target magnet **155**.

[0023] In some cases, coil-over springs may be further added to enhance such resistance. A coil-over spring **170** may be operably coupled to the suspension assembly via an upper spring mounting member and a lower spring mounting member **160**. In some cases, the upper spring mounting member may be the second mechanical interface **146**. In an example embodiment, the lower spring mounting member **160** may be operably coupled to the external body **111**.

[0024] In an example embodiment, the sensor assembly **200** may include an accelerometer **240**. The accelerometer **240** may be a single-axis accelerometer or may be any number of sensors that may accurately determine rate of change of suspension velocity while the vehicle corners or otherwise experiences wheel position changes. In some cases, the accelerometer **240** may be built-in to the linear sensor **220** or may be disposed within the connector assembly **230**. In an example embodiment, the accelerometer **240** may be disposed by the piston head **180** or by the rod **142**. The accelerometer **240** may be disposed anywhere within the sensor assembly **200** and shock absorber **110** that may accurately determine rate of change of suspension velocity during vehicle cornering or other suspension events.

[0025] The accelerometer **240** may determine real-time rate of change of suspension velocity (e.g., along its suspension travel axis), which is the same as or proportional with rate of change of wheel velocity along the longitudinal axis **190**. Thus, for example, the position (and rate of change of velocity) of the wheel is measured between positions of full compression and full rebound. Some other suspension systems may utilize position data determined by the position sensor to determine rate of change of suspension velocity of the suspension system or wheel. To determine such rate of change of suspension velocity, a second derivative would typically need to be taken by the position sensor, thus incorporating a delay with the calculation of rate of change of suspension velocity as well as losing accuracy for the calculations. In this regards, integrating an accelerometer **240** into the sensor assembly **200** or shock absorber **110** may allow for quicker and more accurate data for suspension assembly **100** adjustment and condition determination, as position data and rate of change of suspension velocity may both be provided in real-time locally. In an example embodiment, having accurate, real time position, and rate of change of suspension velocity data measured by the accelerometer **240**, may allow for accurate, real time force per corner measurements.

[0026] In some cases, the rate of change of suspension velocity data from the accelerometer **240** and the displacement data from the linear sensor **220** may be acquired simultaneously and in real time. With simultaneous and real time rate of change of suspension velocity data and displacement

data, suspension stroke position may be determined. Suspension stroke position may be defined by where specifically the suspension assembly is disposed at and the direction the suspension assembly may be headed. Identifying the ride height or suspension assembly **100** position using displacement data from the linear sensor **220** by itself may not fully identify whether the shock absorber **110** may be heading towards maximum compression or maximum extension. Rate of change of suspension velocity data from the accelerometer **240** may identify if the shock absorber **110** may be compressing or extending in the direction of movement represented by the double arrow **165**. In an example embodiment, the maximum compression of the shock absorber may be indicative of the suspension assembly **100** bottoming out and thus not providing the optimum damping. In some cases, the maximum compression of the shock absorber **110** may be circumvented via adjustment of damping of the shock absorber **110** based on the suspension stroke position determination. If the suspension stroke position is determined to be approaching maximum compression, the damping of the shock absorber **110** may be increased. In an example embodiment, the damping of the shock absorber may be adjusted if the suspension stroke position is determined to be approaching minimum compression.

[0027] Furthermore, velocity data may be important for suspension assembly **100** adjustment. With the incorporation of the accelerometer **240** into the shock absorber **110**, the sensor assembly **200** may allow for an alternative method for determining velocity data representative of wheel velocity or vehicle corner velocity. The accelerometer **240** may determine velocity data via the integration of the rate of change of suspension velocity measured. Combined, the velocity data that may be obtained via the first derivative of the position data collected from the linear sensor **220** and the velocity data that may be obtained via the integration of the rate of change of suspension velocity data collected by the accelerometer **240** may create a verified set of velocity data that may be more accurate and help to eliminate/reduce velocity error.

[0028] Referring now to FIGS. 3-6, example embodiments of the suspension assembly **100**, the shock absorber **110**, and the sensor assembly **200** are illustrated. FIG. 3 illustrates a perspective view of a vehicle suspension system in accordance with an example embodiment. FIG. 4 shows an isolated perspective view of a sensor assembly in accordance with an example embodiment. FIG. 5 illustrates an isolated perspective view of a rebound stop for a piston in accordance with an example embodiment. FIG. 6 further shows an isolated perspective view of a shock absorber for a piston in accordance with an example embodiment.

[0029] In some cases, the connector assembly **230** may be the only instance of a connector in the sensor assembly **200**. In this regard, all communication signal transferring, power signal transferring, and grounding within the sensor assembly **200** may be accomplished through the single instance of the connector assembly **230**. In an example embodiment, a Peripheral Sensor Interface 5 (PSI-5) may be used by the sensor assembly **200** and the connector assembly **230** for signal transfer. With PSI-5, the connector assembly **230** may only need a two wire interface to transfer all necessary information and perform the grounding function. The two wire interface may include a power-in wire and a power-out wire, which streamlines the sensor assembly **200** communication compared to older communication methods requiring more complex wiring. In this regard, PSI-5 may allow for the power-out wire of the connector assembly **230** and sensor assembly **200** to output a single data packet for all necessary data. The single data packet may include any necessary instances of displacement data and rate of change of suspension velocity data, streamlining the sensor assembly **200** communication.

[0030] In an example embodiment, the sensor casing **210** may have parallel peripheral edges **211**. The parallel peripheral edges may be disposed substantially parallel to the longitudinal axis **190** of the shock absorber **110**. In some cases, the parallel peripheral edges **211** may include grooves **212**. In this regard, the linear sensor **220** may slide into the grooves **212** of the parallel peripheral edges **211** to operably couple the linear sensor **220** to the sensor casing **210**. In some cases, the linear sensor **220** and the sensor casing **210** may operably couple via a tongue and groove configuration

or via an alternative groove configuration.

[0031] In an example embodiment, the sensor casing **210** may also include a distal support groove **213**. The distal support groove **213** may be substantially perpendicular to the parallel peripheral edges **211**. In some cases, the distal support groove **213** may operably couple to a distal end of the linear sensor **220**. The distal end of the linear sensor **220** may be the end of the linear sensor **220** furthest away from the connector assembly **230**. In an example embodiment, the sensor casing **210** only operably couples to the linear sensor **220** via the parallel peripheral edges **211** and the distal support groove **213**. In some cases, the sensor casing **210** may be a U-shape, formed via the parallel peripheral edges **211** and the distal support groove **213**; however, the sensor casing **210** may be any number of shapes that may operably couple the linear sensor **220** to the shock absorber **110**.

[0032] In an example embodiment, the linear sensor **220** may include a sensor housing **221**. The sensor housing **221** may surround or form an exterior of the linear sensor **220**. The sensor housing **221** may operably couple the linear sensor to the sensor casing **210**.

[0033] In some cases, the suspension assembly **100** may include a sealing member **250**. The sealing member **250** may operably couple to the linear sensor **220**, the sensor casing **210**, and external body **111**. In this regard, the sealing member **250** may create a seal between the linear sensor **220** and the sensor casing **210** to shield the linear sensor **220** from external debris. In some cases, the external debris may be rocks, pebbles, sand, rain, fluid, or any other debris that may be encountered by the vehicle during vehicle operation. In an example embodiment, the sealing member **250** may be a decal or other sticker that is attached via an adhesive provided on a contact side thereof. In some cases, the sealing member **250** may be translucent or transparent in order to see sensor assembly **200** components after application of the sealing member **250**.

[0034] In an example embodiment, the shock absorber **110** may be made of aluminum. In this regard, the material of the shock absorber **110** may not affect inductive detection of the target magnet **155** by the linear sensor **220**. In some cases, the shock absorber **110**, or more specifically the external body **111**, may be any material that does not interfere with inductive detection of the target magnet **155** by the linear sensor **220**.

[0035] In an example embodiment, the lower spring mounting member **160** of the suspension assembly **100** may include a clearance between the external body **111** of the shock absorber **110** and a portion of the lower spring mounting member **160**. In some cases, the components of the sensor assembly **200** (i.e. sensor casing **210** and linear sensor **220**) may be disposed within the clearance and this within the inner circumference of the lower spring mounting member **160**.

[0036] In an example embodiment, the linear sensor **220** may be disposed at, and directly in contact with the external body **111** of the shock absorber **110**, in addition to being operably coupled to the sensor casing **210**. In some cases, the linear sensor **220** may include an adhesive or the sensor housing **221** may include an adhesive to directly affix the linear sensor **220** to the external body **111**. In an example embodiment, the linear sensor **220** and the external body **111** may be secured in contact with one another via welding, an adhesive, or any other method that directly operably couples the linear sensor **220** and the external body **111**. In some cases, the sensor casing **210** may be secured via welding, an adhesive, or any other method that directly operably couples the linear sensor **220** and the external body **111**. In an example embodiment, the linear sensor **220** and the sensor casing may be operably coupled via mounting screws.

[0037] In some cases, the rate of change of suspension velocity data and the displacement data may be utilized to monitor vehicle condition. In an example embodiment, the rate of change of suspension velocity data and the displacement data may assist in determining if the sensor assembly may be in a fault state or if the shock absorber may be in a fault state. In some cases, the rate of change of suspension velocity data and the displacement data may be voltage or current measurements.

[0038] In an example embodiment, the shock absorber **110** may include an external cover that may be disposed over the external body **111** of the shock absorber **110**. In some cases, the external cover

may be operably coupled to the first mechanical interface **144** or the second mechanical interface **146**. In an example embodiment, the external cover may include a sheath to hold the linear sensor **220**. For example, the linear sensor **220** may be operably coupled to the external cover and the target magnet may be operably coupled an alternative component of the suspension assembly to measure displacement data. In some cases, the shock absorber **110** may be a single body shock absorber or a multiple body shock absorber. In an example embodiment, the linear sensor **220** may be over 50% of a longitudinal length of the external body **111** of the shock absorber **110**.

[0039] In an example embodiment, the sensor assembly **200** may be used in semi-active or active suspension assemblies. In some cases, if the sensor assembly **200** is used in an active suspension assembly, the displacement data may allow for a proper imparting force to be applied to the shock absorber **110**.

[0040] In accordance with an example embodiment, a sensor assembly for a suspension assembly of a vehicle may therefore be provided. The sensor assembly may include a sensor for detecting displacement data of a target magnet and a sensor casing operably coupled to an external body of a shock absorber of the suspension assembly and the sensor. The sensor may include a linear flexible printed circuit that may be disposed along a longitudinal axis of the shock absorber to measure displacement data of the target magnet relative to the shock absorber to determine the suspension assembly position. The sensor assembly may further include a single axis accelerometer measuring rate of change of suspension velocity data in a direction along the longitudinal axis of the shock absorber, and the target magnet operably coupled to the suspension assembly via a suspension assembly component other than the external body of the shock absorber.

[0041] The sensor assembly of some embodiments may include additional features, modifications, augmentations and/or the like to achieve further objectives or enhance performance of the device. The additional features, modifications, augmentations and/or the like may be added in any combination with each other. Below is a list of various additional features, modifications, and augmentations that can each be added individually or in any combination with each other. For example, the target may include one or more magnets. In an example embodiment, the rate of change of suspension velocity data and the displacement data may be used simultaneously and in real time to detect suspension stroke position. In some cases, a maximum compression of the shock absorber may be circumvented due to adjustment of damping of the suspension assembly based on the suspension stroke position. In an example embodiment, the suspension assembly component may be a rebound stop of a piston of the shock absorber. In some cases, the sensor only may require a single instance of a connector for communication signal transfer, power signal transfer, and grounding within the sensor assembly. In an example embodiment, a single data packet may be sent through the connector comprising the rate of change of suspension velocity data and the displacement data. In some cases, the sensor casing may further include parallel peripheral edges. The parallel peripheral edges may include grooves. The sensor may slide into the grooves to operably couple the sensor and the sensor casing. In an example embodiment, the sensor casing may only operably couple to the sensor via the parallel peripheral edges and a distal support groove substantially perpendicular to the parallel peripheral edges. The distal support groove may operably couple to a distal end of the sensor. In some cases, a sealing member may operably couple to the sensor, the sensor casing, and the external body of the shock absorber. The sealing member may create a seal between the sensor and the sensor casing from external debris. In an example embodiment, the external body of the shock absorber may be aluminum. In some cases, the sensor may be disposed within an inner circumference of a spring mounting member of the shock absorber. In an example embodiment, the sensor may be disposed at, and in contact with, the external body of the shock absorber.

[0042] In yet another example embodiment, a suspension system for a vehicle may be therefore be provided. The suspension assembly may include a shock absorber operably coupled to a wheel assembly and a chassis of the vehicle, and a sensor assembly operably coupled to the shock

absorber. The sensor assembly may include a sensor for detecting displacement data of a target magnet, a sensor casing operably coupled to an external body of the shock absorber of the suspension assembly and the sensor. The sensor may include a linear flexible printed circuit that may be disposed along a longitudinal axis of the shock absorber to measure displacement data of the target magnet relative to the shock absorber to determine the suspension assembly position. The sensor assembly further include a single axis accelerometer measuring rate of change of suspension velocity data in a direction along the longitudinal axis of the shock absorber, and the target magnet operably coupled to the suspension assembly via a suspension assembly component other than the external body of the shock absorber.

[0043] Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A sensor assembly for a suspension assembly of a vehicle, the sensor assembly comprising, a sensor for detecting displacement data of a target magnet; and a sensor casing operably coupled to an external body of a shock absorber of the suspension assembly and the sensor; wherein the sensor comprises a linear flexible printed circuit disposed along a longitudinal axis of the shock absorber to measure displacement data of the target magnet relative to the shock absorber to determine a suspension assembly position, wherein the sensor assembly further comprises a single axis accelerometer measuring rate of change of suspension velocity data in a direction along the longitudinal axis of the shock absorber, and wherein the target magnet operably coupled to the suspension assembly via a suspension assembly component other than the external body of the shock absorber.
2. The sensor assembly of claim 1, wherein the rate of change of suspension velocity data and the displacement data are used simultaneously and in real time to detect suspension stroke position.
3. The sensor assembly of claim 2, wherein a maximum compression of the shock absorber is circumvented due to adjustment of damping of the suspension assembly based on the suspension stroke position.
4. The sensor assembly of claim 1, wherein the suspension assembly component is a rebound stop of a piston of the shock absorber.
5. The sensor assembly of claim 1, wherein the sensor only requires a single instance of a connector for communication signal transfer, power signal transfer, and grounding within the sensor assembly.
6. The sensor assembly of claim 5, wherein a single data packet is sent through the connector

comprising the rate of change of suspension velocity data and the displacement data.

7. The sensor assembly of claim 1, wherein the sensor casing further comprises parallel peripheral edges, wherein the parallel peripheral edges further comprise grooves, and wherein the sensor slides into the grooves to operably couple the sensor and the sensor casing.

8. The sensor assembly of claim 7, wherein the sensor casing only operably couples to the sensor via the parallel peripheral edges and a distal support groove substantially perpendicular to the parallel peripheral edges, and wherein the distal support groove operably couples to a distal end of the sensor.

9. The sensor assembly of claim 1, wherein a sealing member operably couples to the sensor, the sensor casing, and the external body of the shock absorber, and wherein the sealing member creates a seal between the sensor and the sensor casing from external debris.

10. The sensor assembly of claim 1, wherein the external body of the shock absorber is aluminum.

11. The sensor assembly of claim 1, wherein the sensor is disposed within an inner circumference of a spring mounting member of the shock absorber.

12. The sensor assembly of claim 1, wherein the sensor is disposed at, and in contact with, the external body of the shock absorber.

13. A suspension assembly of a vehicle, the suspension assembly comprising, a shock absorber operably coupled to a wheel assembly and a chassis of the vehicle; and a sensor assembly operably coupled to the shock absorber, the sensor assembly further comprising: a sensor for detecting displacement data of a target magnet; and a sensor casing operably coupled to an external body of the shock absorber of the suspension assembly and the sensor; wherein the sensor comprises a linear flexible printed circuit disposed along a longitudinal axis of the shock absorber to measure displacement data of the target magnet relative to the shock absorber to determine a suspension assembly position, wherein the sensor assembly further comprises a single axis accelerometer measuring rate of change of suspension velocity data in a direction along the longitudinal axis of the shock absorber, and wherein the target magnet operably coupled to the suspension assembly via a suspension assembly component other than the external body of the shock absorber.

14. The suspension assembly of claim 13, wherein the rate of change of suspension velocity data and the displacement data are used simultaneously and in real time to detect suspension stroke position.

15. The suspension assembly of claim 14, wherein a maximum compression of the shock absorber is circumvented due to adjustment of damping of the suspension assembly based on the suspension stroke position.

16. The suspension assembly of claim 13, wherein the suspension assembly component is a rebound stop of a piston of the shock absorber.

17. The suspension assembly of claim 13, wherein the sensor only requires a single instance of a connector for communication signal transfer, power signal transfer, and grounding within the sensor assembly.

18. The suspension assembly of claim 17, wherein a single data packet is sent through the connector comprising the rate of change of suspension velocity data and the displacement data.

19. The suspension assembly of claim 13, wherein the sensor casing further comprises parallel peripheral edges, wherein the parallel peripheral edges further comprise grooves, and wherein the sensor slides into the grooves to operably couple the sensor and the sensor casing.

20. The suspension assembly of claim 19, wherein the sensor casing only operably couples to the sensor via the parallel peripheral edges and a distal support groove substantially perpendicular to the parallel peripheral edges, and wherein the distal support groove operably couples to a distal end of the sensor.
